

ADVANCED JAVA LABSIDDHARTHA ACADEMY OF HIGHER EDUCATION (DEEMED TO BE UNIVERSITY)
(23AI5351) • DEPARTMENT OF AI • BATCH 2025–2029

Week 03: Constructors, Method Overloading & 'this' Keyword

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1. OBJECTIVE / AIM

To write a Java program demonstrating default constructors, parameterized constructors, constructor overloading, method overloading, and the 'this' keyword.

2. CONCEPT & THEORY

Constructors are special member methods called when an instance of an object is created. Method overloading allows multiple methods to share the same name with different parameter types or counts.

3. JAVA SOURCE IMPLEMENTATION

```
// Advanced Java Lab - Experiment 03
// Student: Trivikram | Roll No: 25EU02067
class Box {
    double width, height, depth;

    // Default Constructor
    Box() {
        width = height = depth = 1.0;
    }

    // Parameterized Constructor
    Box(double w, double h, double d) {
        this.width = w;
        this.height = h;
        this.depth = d;
    }

    // Constructor Chaining using this()
    Box(double len) {
        this(len, len, len);
    }

    double volume() {
```

```
        return width * height * depth;
    }
}

public class Week3Demo {
    public static void main(String[] args) {
        Box b1 = new Box();
        Box b2 = new Box(4.0, 5.0, 6.0);
        Box cube = new Box(3.0);

        System.out.println("Default Box Volume: " + b1.volume());
        System.out.println("Custom Box Volume : " + b2.volume());
        System.out.println("Cube Volume      : " + cube.volume());
    }
}
```

4. COMPILATION & EXECUTION OUTPUT

```
Default Box Volume: 1.0
Custom Box Volume : 120.0
Cube Volume      : 27.0
```